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Integrated Port and Vessel Performance Optimization Through AI/ML Driven Analytics: A Case Study from Oil & Gas Company in Indonesia

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Abstract

The maritime logistics sector faces persistent challenges in balancing operational efficiency, cost control, and reliability. In Indonesia's oil and gas downstream distribution, fragmented data systems and limited visibility into port and vessel performance have constrained effective decision-making. With over 80% of Company's hydrocarbon distribution reliant on sea transportation and a supply chain spanning more than 200 terminals across Indonesia, monitoring vessel and port operations is critical. As an Oil & Gas company, needs to improve its agility and competitiveness by developing a real-time visualization and monitoring system for ship and port performance to identify optimization opportunities that reduce in port time, improve vessel utilization, standardize analytics across multiple ports, and support data-driven decision-making to lower logistics costs.

The research results indicated that the framework combined historical and real-time metrics via an integrated data pipeline refining outputs from the existing integrated Logistics Optimization application into predictive and descriptive analytic models. Machine learning algorithms forecast port effectiveness (hours), berth occupancy (%), waiting times (days), and freight cost (USD/KL). Data processing steps include raw data cleaning, benchmark analysis, and performance monitoring with status indicators. Visualization is delivered through interactive dashboards for vessel (ALD, ADD, transit mapping) and port KPIs. Standardized processes were codified across all selected ports, with root-cause and cost-breakdown analyses embedded to ensure operational consistency.

Initial analysis confirmed vessels spend 64% of their lifecycle in port. By targeting a 2.3% reduction in average in port time from 49.47 to 47.12 hours, the model projects annual OPEX savings of USD 17.1 million, while improved vessel scheduling and utilization could unlock an additional USD 7.4 million in value. Dashboards enabled real-time monitoring of key metrics port time, occupancy rates, waiting periods, and freight costs against segmented historical benchmarks, with deviation alerts driving proactive interventions. Observations show that standardizing operational procedures across ports reduces variability in turnaround times by up to 15%, and predictive analytics facilitate a 10% decrease in unexpected berth delays. Conclusions highlight that combining Machine Learning (ML) driven forecasts with process

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standardization not only delivers significant cost savings and utilization gains but also strengthens decision support systems for continual performance improvement.

This work introduces a unified, ML empowered platform that seamlessly integrates vessel and port analytics across multiple locations, delivering the first standardized predictive analytics suite in maritime logistics. It advances the energy industry by enabling real-time, data driven optimization of maritime operations and cost management.

Keywords: port time, shipping cost, marine transportation performance, machine learning

Introduction

Background

As an Oil & Gas Company with multiple roles need to seek solutions for balancing their financial and social performances. Role as a profit generator requires such companies to optimize cost and asset efficiency aspects. Historically, challenges such as fragmented data sources, limited real time visibility, and the absence of a unified performance monitoring framework have hampered efforts to identify inefficiencies and implement targeted improvements. These limitations have not only affected the reliability of port and vessel operations but have also contributed to higher operational expenditure through excessive idle time, suboptimal berth allocation, and inefficient routing decisions.

The Integrated Port & Vessel Performance Optimization (IPVO) Analytics Dashboard was developed address these gaps. By integrating all the operational and commercial data from more than thousand voyages, the dashboard delivers a single, interactive interface that consolidates key performance metrics across ports, vessels, and cargo flows. It leverages both descriptive analytics and AI/ML capabilities, such as port clustering and vessel arrival prediction to provide actionable recommendations, enabling timely interventions at both the operational and strategic levels.

Through enhanced visibility, the IPVO Analytics Dashboard has played a pivotal role in improving operational effectiveness, such as reducing Integrated Port Time (IPT), optimizing berth utilization, and streamlining cargo distribution. Simultaneously, it has contributed to cost reduction by minimizing delays, lowering fuel consumption through informed speed management, and enabling more accura te resource allocation. This paper explores how the IPVO Analytics Dashboard was conceptualized, developed, and deployed, as well as the measurable impact it has generated in O&G Company's downstream maritime logistics operations.

Problem Statement

With the current situation and pain points of the business to operates with best performance, these are the hurdles that must be solved:

1. Opportunity to optimize logistics costs by reducing high port time durations and enhance visibility into vessel activities, as vessels currently spend approximately 64% of their time in port.
2. The current data and performance analysis practices can be further leveraged to generate actionable insights and improve decision-making,
3. Currently there is a need for standardized processes across multiple port locations to improve operational consistency.

Objectives of the study

The objective of this focused study is to answer business pain points through visualizing and monitoring ship and port performance in real-time, including to provide analytics to better analyze and identify potential optimizations for improvements in vessel time proportions and minimize Integrated Port Time.

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These are the business goals that required to be achieved:

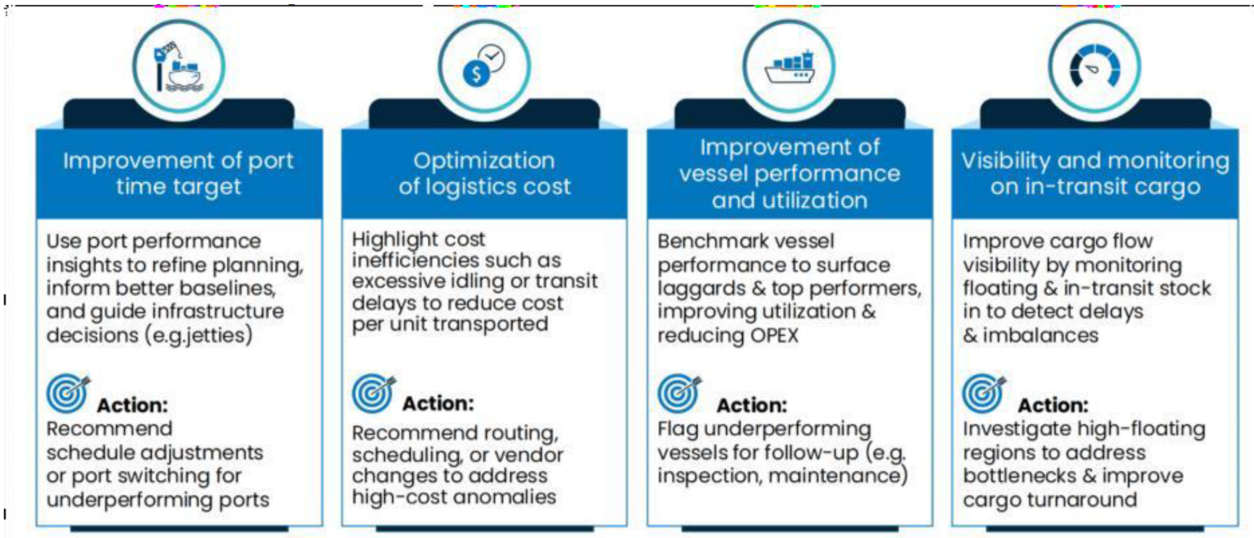

Figure 1—Key Objectives for IPVO

Literature Review

Performance optimization in the maritime logistics industry has been an active area of research, particularly in the domains of port and vessel optimization. The International Maritime Organization's Marine Environment Protection Committee (IMO MEPC, 2010) has been a cornerstone reference in the development of shipping efficiency models. The IMO's guidelines on the Energy Efficiency Design Index (EEDI) and the Ship Energy Efficiency Management Plan (SEEMP) have established internationally recognized metrics for vessel performance evaluation and operational optimization. These models have been widely adopted to predict vessel arrival times and optimize sailing speeds, especially in scenarios where a vessel is predicted to arrive earlier than its allocated Agreed Loading Date (ALD).

More recent studies, such as **Psaraftis and Kontovas (2014)**, explore the trade-offs between fuel efficiency, emissions reduction, and schedule integrity, concluding that dynamic speed optimization, supported by predictive algorithms, can yield substantial economic and environmental benefits. The integration of Artificial Intelligence (AI) and Machine Learning in this context has been demonstrated by **Liu et al. (2021)**, where predictive models improved Estimated Time of Arrival (ETA) accuracy and allowed proactive speed adjustments to optimize port berthing windows.

Port performance measurement has also been extensively studied in the literature. **UNCTAD (2016)** emphasizes that port efficiency is a critical determinant of maritime logistics competitiveness, with vessel turnaround time being a key performance indicator. Similarly, **Cullinane and Wang (2007)** applied Data Envelopment Analysis (DEA) to evaluate port operational efficiency, highlighting that integrated information systems between ports and shipping lines are essential for real-time decision-making.

In terms of integrated optimization platforms, **Stopford (2009)** and **Christiansen et al. (2013)** provide foundational insights into maritime logistics modeling, including vessel routing, scheduling, and cargo allocation optimization. These works stress the importance of incorporating both operational and financial performance indicators in a unified analytics framework—an approach that aligns directly with the objectives of the Integrated Port & Vessel Performance Optimization (IPVO) system.

However, although the aforementioned frameworks have been widely implemented to manage port and vessel performance, the review process confirms that Company's business processes grounded in the IMO's regulatory framework, slow steaming economics, port efficiency measurement, integrated vessel scheduling models, and AI enhanced predictive analytics provide a robust foundation for IPVO. This integrated

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analytics dashboard bridges operational, environmental, and financial dimensions, enabling decision makers to systematically enhance port and vessel performance within the broader maritime logistics network.

Methodology

This research employed literature review and use case method to build a Port and Vessel Optimization Analytics Dashboard in O&G Company. The development and implementation of the Integrated Port & Vessel Performance Optimization (IPVO) Analytics Dashboard followed a structured, multiphase methodology designed to ensure technical robustness, operational relevance, and stakeholder adoption. The approach combined data integration, analytics design, AI/ML development, and iterative validation with end- users.

Project Scope Definition

The Integrated Port & Vessel Performance Optimization (IPVO) project was conceived as a strategic initiative to enhance the operational efficiency, reliability, and cost effectiveness of O&G's maritime logistics network. The system is designed to integrate multi-source operational data into a unified analytics environment, enabling real time monitoring, predictive insights, and data-driven decision making. The project scope has evolved over time, beginning with a clearly defined Initial Scope and subsequently expanding into a more advanced Expanded Scope to address additional optimization opportunities.

Initial Scope. The initial phase of the IPVO project focused on establishing a robust analytical foundation to address three primary operational domains:

i. Analysis of Port Performance

   This component involved the systematic evaluation of port operational efficiency using key performance indicators (KPIs) such as berth occupancy rates, vessel turnaround times, cargo handling productivity, and port congestion levels. The goal was to identify bottlenecks, uncover performance trends, and support informed decision making for capacity planning and process improvement.

ii. Analysis of Vessel Performance

   Vessel performance monitoring was incorporated to track speed profiles, bunker consumption, schedule adherence, and utilization rates. The analysis provided insights into operational efficiency, compliance with optimal sailing speeds, and adherence to planned voyages. This data-driven approach aimed to reduce fuel costs, minimize delays, and ensure alignment with environmental regulations.

iii. Cargo Monitoring

   Real time cargo tracking and monitoring were integrated to oversee the movement, volume, and type of cargo handled at each port. The system provided visibility into cargo allocation efficiency, load/discharge timelines, and cargo flow consistency across the maritime network, supporting better coordination between port and vessel operations.

To operationalize these analytical capabilities, the Initial Scope included the development of 18 Power BI dashboard pages. Each dashboard was purpose built to cover specific analysis areas, offering both summary level overviews and drill-down capabilities into granular operational details. This structured visualization framework enabled stakeholders to monitor operational health, compare performance across assets, and identify actionable improvement areas.

Expanded Scope. Following the successful implementation of the initial phase, the IPVO project scope was broadened to incorporate predictive analytics and advanced decision-support tools. This expansion addressed the need for proactive operations management and more sophisticated optimization strategies.

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i. Implementation of a Port Clustering Model

A data-driven Port Clustering model was introduced to categorize ports based on their structural and operational characteristics, including berth length, storage capacity, crane availability, channel depth, and historical throughput. This clustering approach allowed for more targeted operational strategies, improved resource allocation, and tailored performance benchmarks for ports with similar profiles.

ii. Deployment of a Vessel Early/Late Arrival Prediction Model

The project integrated an Arrival Prediction model leveraging historical voyage data, weather patterns, port congestion statistics, and vessel speed profiles to forecast deviations from scheduled arrival times. This predictive capability enabled stakeholders to anticipate early or late arrivals and take corrective actions such as adjusting sailing speeds or resequencing port calls well in advance of potential schedule disruptions.

iii. Addition of an ALD Alert System

To further strengthen schedule adherence, the ALD Alert System was deployed. This feature automatically notifies relevant stakeholders when predicted arrival dates deviate from the agreed loading date schedules. Alerts are designed to prompt proactive discussions and operational adjustments, minimizing the risk of contractual penalties, idle waiting times, or resource misalignments at the port.

Strategic Value of the Defined Scope. By progressing from the Initial Scope to the Expanded Scope, the IPVO project has transitioned from a primarily descriptive analytics platform to a fully integrated operational intelligence system. The solution now combines historical analysis, real-time monitoring, and predictive modeling to optimize both port and vessel performance while aligning with O&G Company's strategic goals of operational excellence, cost efficiency, and environmental compliance.

This phased scope definition not only ensures that foundational capabilities are firmly established before introducing advanced functionalities but also enables a scalable framework that can accommodate future enhancements such as AI- driven berth scheduling, real time weather routing integration, and automated performance benchmarking across the maritime supply chain.

Data Integration and Pipeline Development

A critical enabler of the Integrated Port & Vessel Performance Optimization (IPVO) system is the seamless integration of heterogeneous data sources into a unified analytics environment. Given the complexity of Company's maritime logistics operations, the success of IPVO depended heavily on establishing a robust data pipeline capable of consolidating, transforming, and validating data from multiple platforms with varying structures, timeframes, and quality standards.

Primary Data Sources. Two primary data repositories were identified as the foundation of the IPVO analytical framework:

i. Integrated Logistics Optimization (ILO) System

The Integrated Logistics Optimization (ILO) application serves as a critical digital platform for monitoring and evaluating the performance of both ports and vessels within maritime supply chain. By consolidating multiple operational metrics into a single interface, ILO enables comprehensive visibility, data driven decision making, and proactive efficiency improvements across the logistics network.

One of the key features of ILO is its ability to capture waiting time per voyage, which measures the duration vessels spend idling before berthing. This metric highlights potential inefficiencies in scheduling, berth allocation, or port congestion, and provides a clear basis for optimizing vessel arrival planning. Complementing this is the Integrated Port Time function, which records the

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complete time cycle of a vessel within port operations from arrival and berthing to cargo handling and departure thereby offering a holistic view of port turnaround performance.

ILO also incorporates real-time data input, allowing operational teams to update and track vessel activities as they occur. This live data stream enhances situational awareness and ensures that any delays, anomalies, or deviations are immediately visible to stakeholders. Through this capability, decision makers can evaluate time effectiveness by comparing actual operational durations against planned schedules and performance benchmarks.

Another important element tracked by ILO is the Slow Pumping Vessel indicator, which identifies vessels with lower than expected cargo discharge or loading rates. Detecting such inefficiencies is crucial for minimizing port stay durations, reducing demurrage risks, and maintaining high levels of operational productivity. In parallel, the system provides detailed insights into shipping costs, linking operational efficiency with financial performance. By analyzing bunker consumption, waiting times, and pumping rates, ILO helps establish a direct relationship between operational effectiveness and cost competitiveness.

Taken together, these features make ILO a comprehensive tool for assessing port and vessel performance. By integrating voyage waiting times, port turnaround metrics, real time operational data, time effectiveness measures, vessel pumping efficiency, and shipping costs, the application empowers Company to identify inefficiencies, implement targeted improvements, and achieve higher levels of reliability and cost optimization across its maritime logistics operations.

ii. Integrated Port Management (IPMAN) System

The Integrated Port Management (IPMAN) System was developed as a centralized platform to coordinate and oversee the operational activities of port facilities. Its primary function is to streamline port management processes by integrating scheduling, berthing allocation, and vessel monitoring into a single, unified system. Through this application, port authorities and logistics teams are able to ensure that resources are utilized efficiently, operational bottlenecks are minimized, and overall service reliability is improved.

A core capability of IPMAN lies in its ability to monitor and manage the waiting status of vessels at anchorage or berth. By systematically recording the duration of waiting times and correlating them with berth availability and port activity levels, IPMAN provides visibility into one of the most critical indicators of port performance. This functionality allows operators to identify congestion trends, anticipate delays, and optimize berth allocation strategies to minimize idle time.

In addition to tracking vessel waiting status, IPMAN serves as a management tool for port operations more broadly. It captures data on vessel arrivals and departures, port occupancy rates, and turnaround times, thereby creating a detailed operational record that supports both day to day management and long-term planning. The system also enables coordination across different port functions such as cargo handling, fueling, and maintenance activities ensuring that operations are synchronized and executed with greater time efficiency.

By consolidating operational data into a structured and accessible format, IPMAN contributes significantly to enhancing port performance and service levels. It provides stakeholders with the information required to make informed decisions, reduces uncertainty in vessel scheduling, and strengthens coordination between shipping lines and port operators. Ultimately, the IPMAN System lays the groundwork for more advanced digital platforms, such as the Integrated Port & Vessel Performance Optimization (IPVO) initiative, by serving as a reliable repository of historical records and operational insights.

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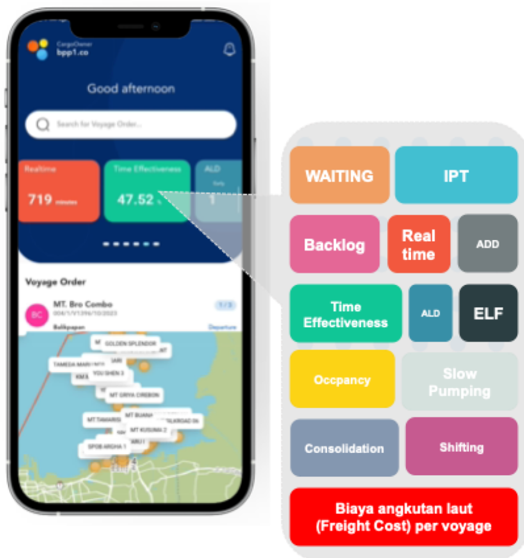

Figure 2—The Menu of Integrated Logistics Optimization (ILO) Apps

Integration Process. The integration of these two data environments required a carefully structured, multi-phase process to ensure data consistency, integrity, and usability within the IPVO dashboards.

i. Data Colocation and Cross Functional Coordination

The first step in the integration process involved data colocation between the Logistics Optimization (LO) team, the Enterprise Information Technology (EIT) team, and the Data Foundation (DF) team. This cross-functional collaboration ensured that data from the ILO and IPMAN platforms could be accessed, compared, and mapped against a shared schema. By establishing a common data dictionary and mapping rules, the teams mitigated inconsistencies in field naming conventions, measurement units, and timestamp formats.

ii. Transformation and Normalization of Datasets

Once data colocation was achieved, the next stage focused on data transformation and normalization. This step was essential for aligning Key Performance Indicators (KPIs) across systems, ensuring that metrics such as vessel turnaround time, berth occupancy rates, bunker consumption, and schedule adherence were calculated consistently. Transformation rules addressed discrepancies in raw data structures, converted units of measure where necessary, and standardized categorical values to enable accurate aggregation and comparison across the integrated dataset.

iii. Data Validation and Anomaly Detection

Before ingestion into the IPVO dashboards, the integrated datasets underwent rigorous validation processes to detect and correct anomalies. Validation checks included:

a. Identifying outliers in operational metrics (e.g., abnormally short or long port stays)
b. Resolving data duplication issues across the two source systems
c. Reconciling mismatched records by cross-referencing timestamps, vessel identifiers, and cargo details

Automated validation scripts were supplemented by manual review from subject matter experts, ensuring that domain knowledge informed the resolution of complex discrepancies. This multi-layered approach safeguarded the accuracy of the analytical outputs and maintained user confidence in the IPVO platform.

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Analytics and Dashboard Design

The IPVO analytics and dashboard design was built upon a structured data pipeline that consolidates inputs from both legacy and modern operational systems. As illustrated in the figure, the architecture integrates two primary sources of maritime logistics data the Integrated Logistics Optimization (ILO) application, which delivers near real time operational data, and the Integrated Port Management (IPMAN) system, which provides historical port operation records.

Automated and Real Time Data Flow. Based on Fig-3 below, The ILO system connects to the ODP Data Lake and feeds into the Data Warehouse via an automated pipeline that refreshes every two hours. This ensures that performance dashboards reflect up-to-date vessel movements, port activity, cargo operations, and time-effectiveness metrics. The continuous refresh cycle is critical for operational decision-making, as it allows users to track disruptions, monitor schedule adherence, and respond quickly to emerging bottlenecks.

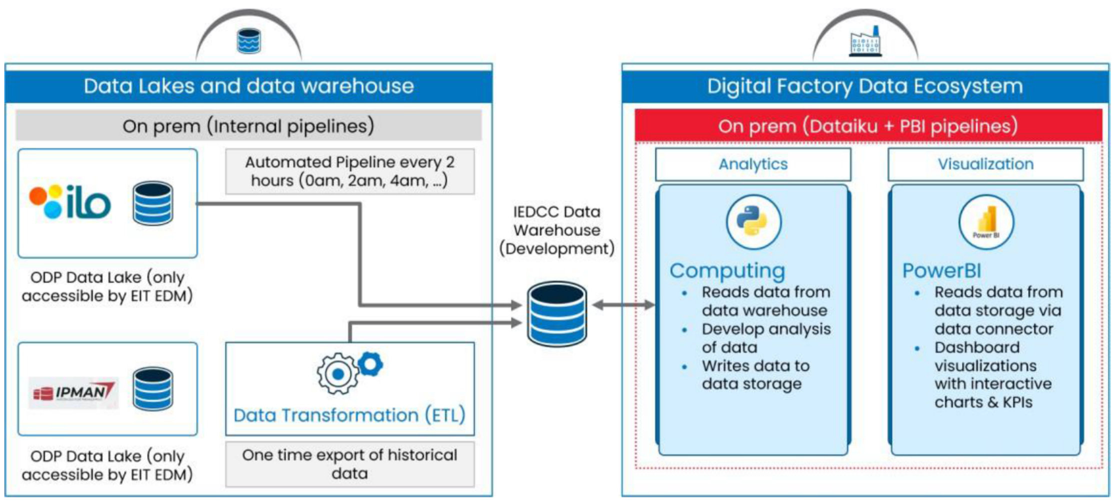

Figure 3—ILO and IPMAN Data Sources, Pipeline, and Visualization

Historical Data Integration. In contrast, the IPMAN platform—positioned as a legacy system—does not provide continuous automated feeds. Instead, its contribution to the IPVO environment occurs through a one-time export of historical data, which is then processed through data transformation (ETL) procedures. These transformations standardize data formats, align KPIs across systems, and remove inconsistencies prior to ingestion into the IEDCC Data Warehouse. Although IPMAN data is static, it provides valuable historical benchmarks that enrich long-term performance analysis and predictive modeling.

Data Transformation and Validation. The pipeline design incorporates a data transformation and validation layer to ensure reliability before dashboard consumption. This process addresses anomalies such as mismatched timestamps, duplicated voyage records, or incomplete cargo handling logs. By harmonizing operational and historical data, IPVO establishes a single source of truth for port and vessel performance metrics.

Dashboard Design and Insight. The output of this data pipeline is visualized through a suite of Power BI dashboards that translate complex datasets into actionable insights. The dashboards are structured to support three analytical tiers:

i. Operational Monitoring
   Provides near real-time indicators such as vessel waiting time, port turnaround time, slow pumping performance, and berth utilization.
ii. Performance Analysis

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Compares operational outcomes against historical benchmarks derived from IPMAN data, enabling stakeholders to identify efficiency gaps.

iii. Predictive and Prescriptive Insights

Integrates predictive models such as early/late arrival forecasts and ALD alerts, supporting proactive decision- making and schedule optimization.

AI/ML Component Development

A distinguishing feature of the Integrated Port & Vessel Performance Optimization (IPVO) project is the embedding of Artificial Intelligence (AI) and Machine Learning (ML) models directly into the analytics dashboard. These models extend the platform beyond descriptive analytics into predictive and prescriptive domains, providing stakeholders with actionable insights that support both day-to-day operational decisions and long-term strategic planning. Two key models have been deployed within the IPVO framework: the Port Clustering Model and the Vessel Arrival Prediction Model.

Port Clustering Model. The Port Clustering Model leverages K-means clustering combined with dimensionality reduction techniques to classify ports into homogeneous groups based on structural and operational attributes. Key input features include jetty count, storage capacity, annual throughput, and the types of vessels serviced. By grouping ports with similar profiles, the model enables apples-to-apples KPI benchmarking, ensuring that performance comparisons are meaningful and contextually valid.

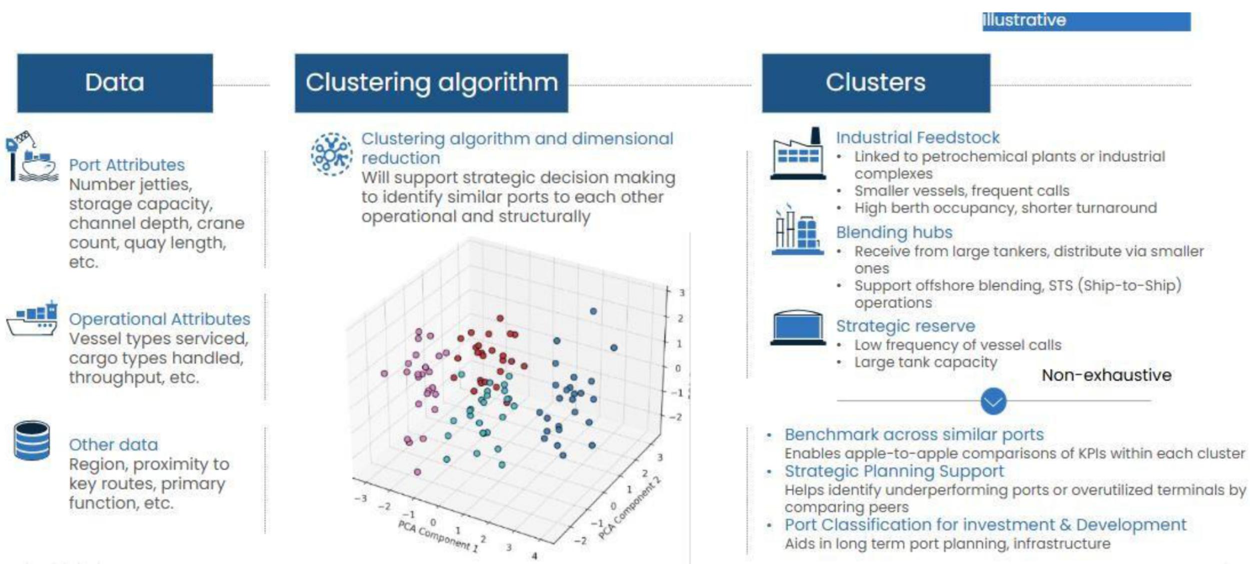

Figure 4—Powering Port analytics and benchmarking through Smart Clustering

This clustering capability also provides valuable insights for long-term infrastructure planning, as it highlights underperforming ports within each cluster and identifies structural characteristics associated with higher levels of efficiency. For Pertamina, this model offers a data-driven foundation for prioritizing investment decisions, capacity expansion, and resource allocation across its port network.

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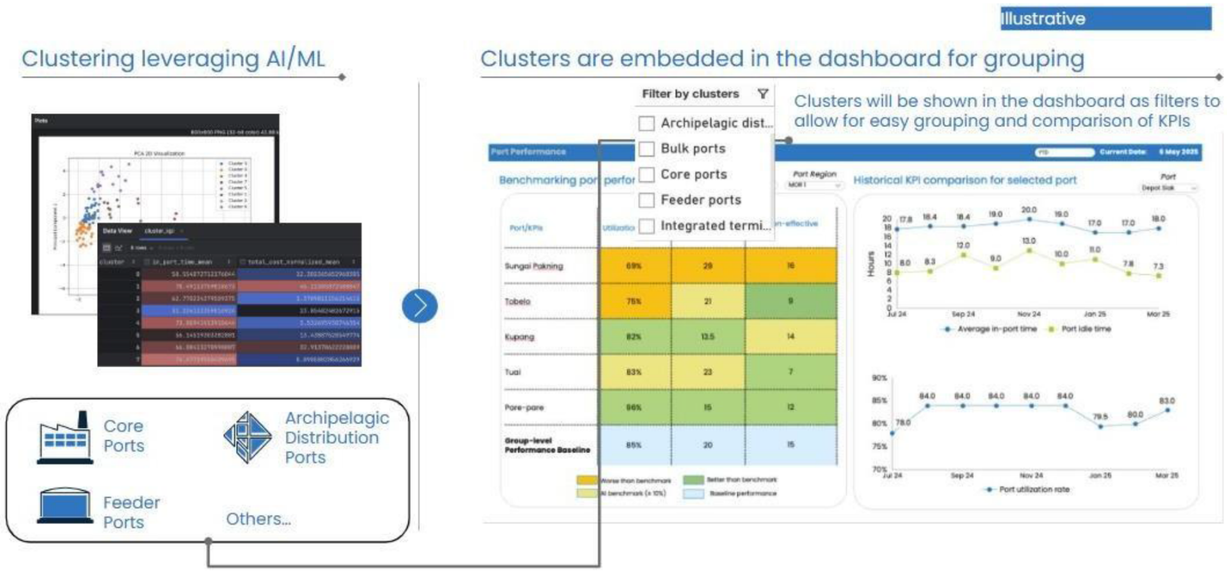

Figure 5—Port clustering results to improve the apple-to-apple comparison

Vessel Prediction Model. The Vessel Arrival Prediction Model applies a combination of regression techniques and time-series analysis to forecast deviations between planned and actual arrival times. Inputs to the model include historical voyage data, vessel speed profiles, port congestion levels, and schedule records. By predicting early or late arrivals in advance, the system provides operational teams with the opportunity to implement early interventions such as adjusting sailing speeds, resequencing berth assignments, or reallocating port resources.

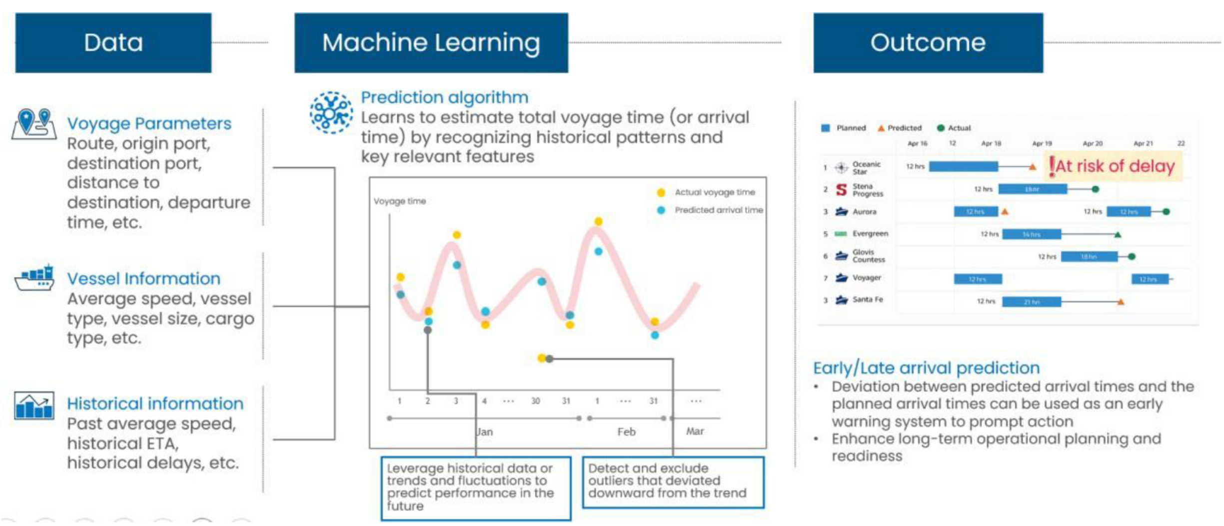

Figure 6—Leveraging Predictive Machine Learning to Prompt action on Early/Late Arrival

This predictive capability is particularly important in preventing port congestion and idle waiting times, which are critical drivers of both operational cost and service reliability. By proactively flagging schedule risks, the model ensures that port and vessel operations remain synchronized with planned logistics flows.

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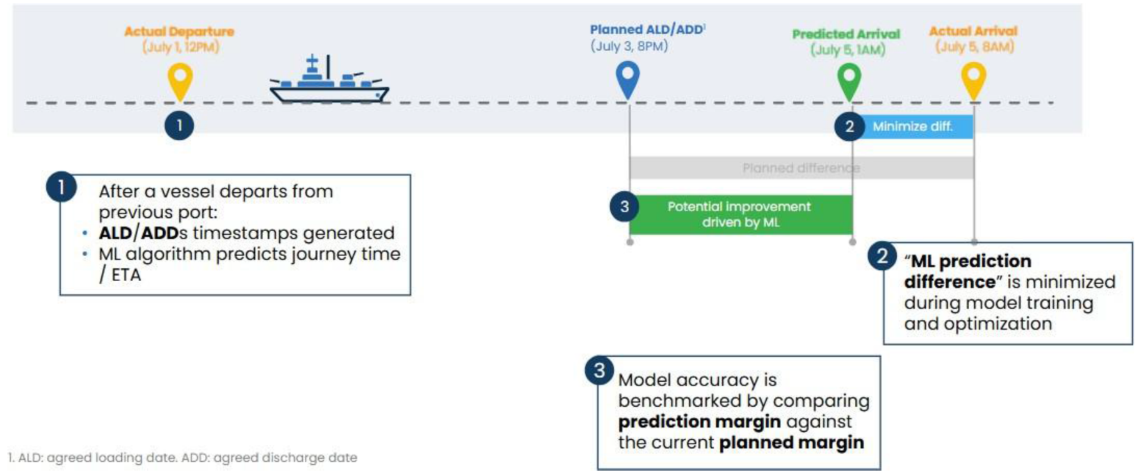

Figure 7—Illustration of How the Model Works for Late Arrival

Strategic Contribution of AI/ML in IPVO. Together, these two models illustrate how IPVO integrates advanced analytics into its dashboard environment to go beyond monitoring and reporting. The Port Clustering Model strengthens strategic decision making by aligning port performance with infrastructure development goals, while the Vessel Arrival Prediction Model enhances operational reliability by minimizing schedule disruptions.

The combined effect of these AI/ML models is a platform that not only reports performance but also anticipates challenges, prescribes solutions, and supports data-driven decision making across Company's maritime logistics ecosystem.

Validation and Iterative Refinement

To ensure the reliability, usability, and operational relevance of the Integrated Port & Vessel Performance Optimization (IPVO) platform, the dashboard and embedded AI/ML models underwent a structured process of validation and iterative refinement. This process was designed to balance technical robustness with practical applicability by actively involving end-users in testing and feedback cycles. The validation framework was carried out over multiple development sprints, enabling continuous improvement and alignment with operational requirements.

Prototype Demonstrations for Stakeholder Review. Early prototypes of the dashboards and predictive models were presented to key stakeholders including logistics planners, port operators, and vessel scheduling teams for review and evaluation. These demonstrations allowed stakeholders to provide direct input on the design, functionality, and analytical outputs, ensuring that the platform addressed actual operational pain points rather than theoretical objectives.

Scenario Testing with Historical Live Data. The validation process incorporated scenario-based testing, leveraging both historical datasets and live operational data streams. Historical data from the IPMAN platform was used to benchmark the accuracy of predictive models, while real- time data from the ILO system was applied to evaluate system responsiveness under live operational conditions. This dual testing approach verified that the models not only replicated past performance patterns accurately but also adapted effectively to current operational dynamics.

UI/UX Refinements for Enhanced Interpretability. Feedback from stakeholders emphasized the need for dashboards that were not only analytically powerful but also intuitive and fast to interpret in high-pressure decision-making environments. As a result, iterative UI/UX refinements were incorporated, focusing on visual clarity, logical navigation, and interactive features such as drill-down capabilities. These refinements

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reduced cognitive load for users and improved the speed at which insights could be translated into operational actions.

Result and Discussion

By doing this work as a unified, AI/ML empowered analytics that seamlessly integrates vessel and port performance data, company can have clear recommendations to be taken into action and leverage the opportunity of cost breakdown, port & vessel efficiency and effectiveness

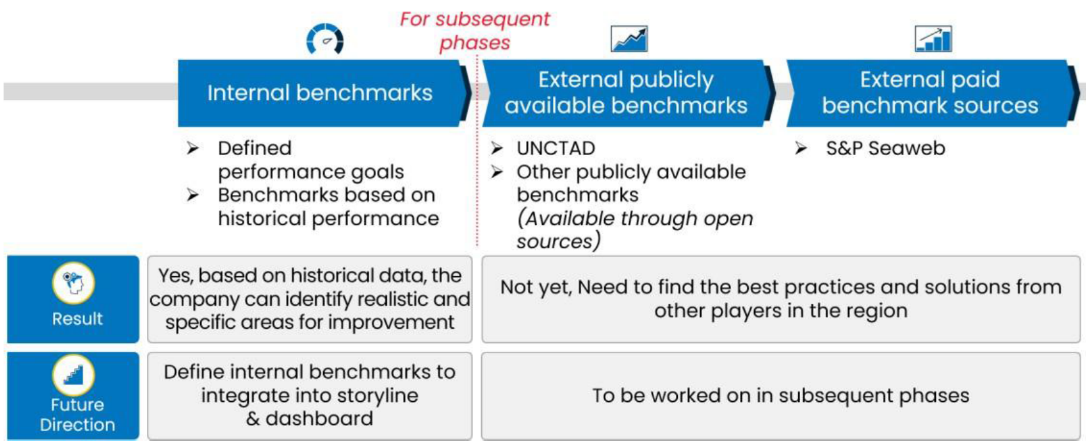

Figure 8—Internal Benchmark to Enable Actionable Insight

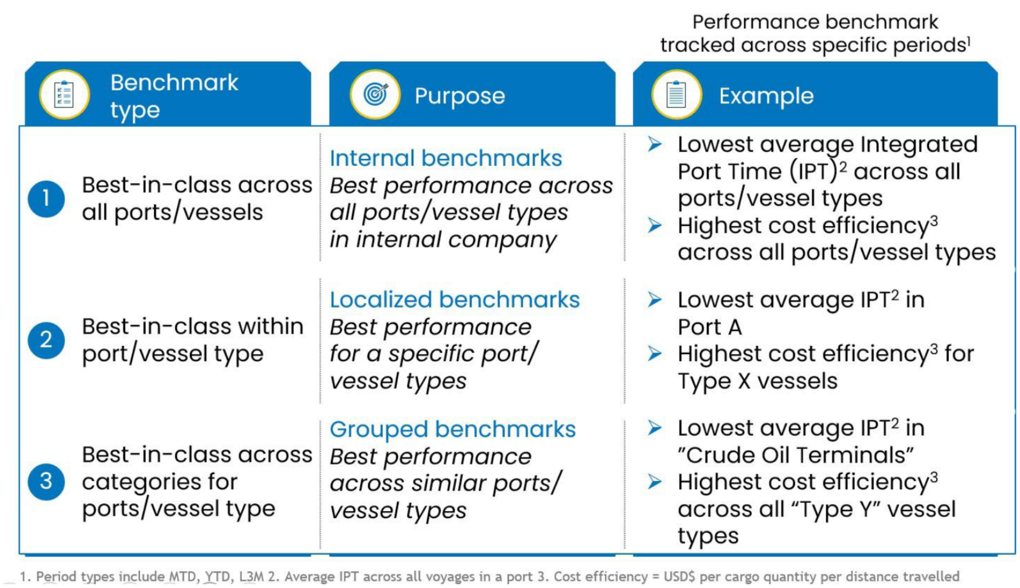

Figure 9—Defining Benchmark Based on Historical Performance

As a result of port clustering methodology, company's integrated port time achievement was reduced by 6% to 46.63 hours in June 2025. The condition needs to be maintained and improve by validating IPMAN historical data & ILO operations data (data quality) also enhancing modelling features using richer data and model re-training.

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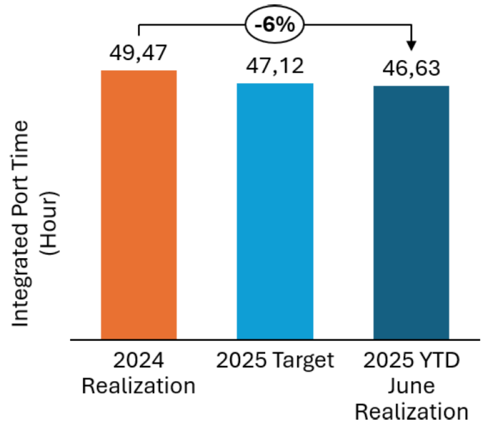

Figure 10—Internal Benchmark Comparation Result Based on Historical Performance

Conclusion and Suggestions for Future Work

The Integrated Port & Vessel Performance Optimization (IPVO) Analytics Dashboard has proven to be a transformative tool for Oil & Gas Company's downstream maritime logistics operations. By consolidating fragmented data sources from the ILO and IPMAN systems, integrating AI/ML driven analytics, and delivering real time performance visibility, the dashboard has enabled both tactical and strategic improvements.

Operationally, IPVO can reduce Integrated Port Time, optimized berth utilization, minimized idle fuel consumption, and improved cargo distribution efficiency. Strategically, the implementation of port clustering and vessel arrival prediction models has empowered decision- makers to benchmark performance, plan infrastructure investments, and proactively manage operational risks.

The initiative not only delivered tangible cost savings but also established a scalable digital framework capable of evolving with Company's operational needs. The combination of descriptive analytics, predictive modeling, and actionable recommendations has shifted decision making from reactive to proactive supporting Company's broader goal of enhancing supply chain resilience and operational excellence.

While the current implementation has demonstrated significant benefits, further enhancements can unlock additional value:

A. Integrate External Data Sources

   Incorporate AIS (Automatic Identification System) feeds, weather forecasts, and port authority schedules to enhance predictive accuracy.

B. Advanced Predictive Models

   Develop and deploy models for fuel consumption forecasting, maintenance prediction, and dynamic berth scheduling.

C. Automated Decision Support

   Implement prescriptive analytics to recommend and automatically trigger operational adjustments (e.g., rescheduling, routing changes) based on AI/ML predictions.

By pursuing these enhancements, Company can further strengthen the IPVO as a cornerstone of its digital logistics strategy driving sustained efficiency gains, cost reductions, and supply chain reliability in the years ahead.

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